Light

# GCSE Computer Science — Programming Fundamentals (Set 2)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

## Q1 (1 mark)

Consider the following line of code:

```
score = 50
```

What is the term for giving a variable its first value?

- ○ **A.** Initialisation
- ○ **B.** Compilation
- ○ **C.** Debugging
- ○ **D.** Interpretation

## Q2 (1 mark)

A programmer wants to store a student's total marks in a variable.
Which variable name follows good programming practice?

- ○ **A.** x
- ○ **B.** 123total
- ○ **C.** total marks
- ○ **D.** total_marks

Q3 (1 mark)

Consider the following assignment:

```
message = "Hello World"
```

What is the data type of the variable `message`?

- ○ **A.** Integer
- ○ **B.** Boolean
- ○ **C.** String
- ○ **D.** Real

Q4 (1 mark)

A variable needs to store the value 3.14159 representing pi.
Which data type should be used?

- ○ **A.** Real (float)
- ○ **B.** Integer

- ○ **C.** Boolean
- ○ **D.** String

Q5 (1 mark)

Consider the following code:

```
result = 6 * 7
```

What value is stored in `result`?

- ○ **A.** 42
- ○ **B.** 13
- ○ **C.** 67
- ○ **D.** 1

Q6 (1 mark)

Consider the following code:

```
answer = 15 / 3
```

What value is stored in `answer`?

- ○ **A.** 5
- ○ **B.** 5.0
- ○ **C.** 45

- ○ **D.** 12

Q7 (1 mark)

Consider the following code:

```
power = 2 ** 4
```

What value is stored in `power`?

- ○ **A.** 16
- ○ **B.** 8
- ○ **C.** 6
- ○ **D.** 24

Q8 (1 mark)

Consider the following expression:

```
10 < 5
```

What value does this expression evaluate to?

- ○ **A.** False
- ○ **B.** True
- ○ **C.** 5
- ○ **D.** 10

Q9 (1 mark)

Consider the following expression:

```
7 != 7
```

What value does this expression evaluate to?

- A. True
- B. False
- C. 0
- D. 14

Q10 (1 mark)

Consider the following expression where `x = 3`:

```
x > 10 or x < 5
```

What value does this expression evaluate to?

- A. False
- B. True
- C. 3
- D. Error

Q11 (1 mark)

What is the purpose of the `print()` function in Python?

- ○ **A.** To display output to the screen
- ○ **B.** To store data in a variable
- ○ **C.** To read input from the keyboard
- ○ **D.** To perform calculations

Q12 (1 mark)

A loop continues running forever because its terminating condition is never met.
What is this situation called?

- ○ **A.** A nested loop
- ○ **B.** A for loop
- ○ **C.** An infinite loop
- ○ **D.** A counter loop

Q13 (2 marks)

Consider the following code:

```
age = int("25")
print(age + 5)
```

What is output by this code?

- ○ **A.** 255
- ○ **B.** 30
- ○ **C.** "25" + 5

- ○ **D.** Error

Q14 (2 marks)

Consider the following code:

```
count = 10
message = "Items: " + str(count)
print(message)
```

What is output by this code?

- ○ **A.** Items: 10
- ○ **B.** Items: count
- ○ **C.** Error
- ○ **D.** Items: + count

Q15 (2 marks)

Consider the following code:

```
grade = 65
if grade >= 70:
    result = "A"
elif grade >= 60:
    result = "B"
elif grade >= 50:
    result = "C"
else:
    result = "F"
print(result)
```

What is output by this code?

- ○ **A.** A
- ○ **B.** B
- ○ **C.** C
- ○ **D.** F

Q16 (2 marks)

Consider the following code:

```
x = 1
while x <= 3:
    print(x)
    x = x + 1
```

What is output by this code?

- ○ **A.** 1 2 3 4 (each on a new line)
- ○ **B.** 1 2 3 (each on a new line)
- ○ **C.** 0 1 2 3 (each on a new line)
- ○ **D.** Nothing is output

Q17 (2 marks)

Consider the following code:

```
for i in range(5, 8):
    print(i)
```

What is output by this code?

- ○ **A.** 5 6 7 8 (each on a new line)
- ○ **B.** 0 1 2 3 4 5 6 7 (each on a new line)
- ○ **C.** 5 6 7 (each on a new line)
- ○ **D.** 8 (once)

Q18 (2 marks)

What is a procedure in programming?

- ○ **A.** A variable that stores multiple values
- ○ **B.** A type of loop that repeats a fixed number of times
- ○ **C.** A named block of reusable code that performs a task but does not return a value
- ○ **D.** An error that occurs during program execution

Q19 (2 marks)

Consider the following code:

```
def greet(name):
    print("Hello, " + name)

greet("Sam")
```

What is the relationship between `name` and `"Sam"`?

- ○ **A.** name is an argument; "Sam" is the parameter
- ○ **B.** Both are parameters

- ○ **C.** Both are arguments
- ○ **D.** name is a parameter; "Sam" is the argument passed to it

Q20 (2 marks)

Consider the following code:

```
def double(n):
    return n * 2

result = double(5)
print(result)
```

What is output by this code?

- ○ **A.** 5
- ○ **B.** n * 2
- ○ **C.** 10
- ○ **D.** double(5)

Q21 (2 marks)

A program calculates the average of three test scores but always produces a result that is too high.
What type of error is this most likely to be?

- ○ **A.** Syntax error
- ○ **B.** Logic error
- ○ **C.** Runtime error

- D. Compilation error

Q22 (2 marks)

Consider the following code:

```
word = "Python"
print(word[0])
```

What is output by this code?

- A. y
- B. Python
- C. 0
- D. P

Q23 (2 marks)

Consider the following expression where `a = 5` and `b = 10`:

```
a >= 5 and b <= 10
```

What value does this expression evaluate to?

- A. False
- B. 5
- C. True
- D. 10

Q24 (3 marks)

Consider the following code:

```
for i in range(2):
    for j in range(3):
        print(i, j)
```

How many times is the print statement executed?

- A. 5
- B. 3
- C. 2
- D. 6

Q25 (3 marks)

Consider the following code designed to swap two values:

```
a = 5
b = 10
temp = a
a = b
b = temp
print(a, b)
```

What is output by this code?

- A. 5 10
- B. 10 10
- C. 10 5

- ○ **D.** 5 5

Q26 (3 marks)

Consider the following code:

```
total = 0
for num in [2, 4, 6]:
    total = total + num
print(total)
```

What is output by this code?

- ○ **A.** 6
- ○ **B.** 246
- ○ **C.** 0
- ○ **D.** 12

Q27 (3 marks)

Consider the following code:

```
def add(a, b):
    return a + b

def multiply(x, y):
    return x * y

result = multiply(add(2, 3), 4)
print(result)
```

What is output by this code?

- A. 9
- B. 14
- C. 20
- D. 24

Q28 (3 marks)

Consider the following code:

```
colours = ["red", "green", "blue"]
print(colours[3])
```

What happens when this code runs?

- A. Outputs "blue"
- B. Outputs nothing
- C. Syntax error
- D. Runtime error (IndexError)

Q29 (3 marks)

Consider the following code:

```
def calculate():
    value = 100
    return value

value = 50
answer = calculate()
print(value)
```

What is output by this code?

- ○ **A.** 100
- ○ **B.** 50
- ○ **C.** Error
- ○ **D.** 150

Q30 (3 marks)

A programmer adds the following code before processing user input:

```
if age < 0 or age > 150:
    print("Invalid age entered")
else:
    # process the age
```

What programming practice does this demonstrate?

- ○ **A.** Code commenting
- ○ **B.** Variable initialisation
- ○ **C.** Loop optimisation
- ○ **D.** Defensive programming with input validation

Total: **55** marks • Time: **30 min**

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